Stellar Structure Calculation for a 5 $M_\odot$ ZAMS Star

Leonello Castro Cillis[1, 2]

[1]Department of Physics and Astronomy, Johns Hopkins University, Baltimore, MD 21218, USA

[2]Space Telescope Science Institute, 3700 San Martin Drive, Baltimore, MD 21218, USA

ABSTRACT

In this project, we analyze the stellar interior of a Zero Age Main Sequence (ZAMS) Star. We start from the four coupled differential equations of stellar evolution that describe the relationships between the stellar variables (luminosity, pressure, radius, and temperature) with respect to the Lagrangian coordinate, which is the enclosed mass from the star's core. We develop a Python routine to propagate through these differential equations in two ways: one starts from the boundary conditions in the star's core and the other in the star's surface. The solution propagates to the point where half the stellar mass is reached, and we repeat this process until the stellar variables from both solutions are very close to each other. We present in this report the results for a 5 $M_\odot$ star with a composition of $X = 0.70$, $Y = 0.28$, $Z = 0.02$, although the code is fit to run a much larger range of masses and compositions as well. We find that it takes 19 iterations to achieve a converged result. We present which regions of the star are convective and radiative. We find that the energy transport mechanism is convective up until the enclosed mass reaches 0.2312 M_*, after which the energy transport becomes radiative (convective core, radiative envelope). We determine the luminosity, radius, effective temperature, and gravity at the surface of the star. We then compare our results with MESA (Modules for Experiments in Stellar Astrophysics). We find reasonable agreement between all the stellar variables: the stellar radius, surface gravity, and effective temperature fall to within 10% of their respective values from MESA, and the luminosity falls to within 25% of its value from MESA.

We invoke multiple assumptions and constraints in our code. First, we recognize that this is a ZAMS star, so no other phases of the star's evolution are considered. We also neglect the effects of the star's rotation and magnetic field. We assume that Local Thermodynamic Equilibrium applies. Finally, we assume the gas is fully ionized and a homogeneous chemical mixture of Hydrogen, Helium, and metals.

For more information on how to run this code, along with the results from MESA, we have made the code publicly available for this project on GitHub[1].

1. OVERVIEW OF STELLAR MODELS

The physics behind stellar structure and stellar evolution has been a complicated field of study for many astronomers throughout the years, particularly because the relevant properties of a star are not usually found through pure observation. Instead, scientists often create numerical models based on well understood physical properties. As such, many models that were created at a certain time have often been revisited and reconstructed to account for new physical considerations that could be put into effect into those models. These can be changes to the physical assumptions or changes due to improvements to the computing power and numerical techniques to further advance those models (Hansen et al. 2004).

The most important parameter to understanding a star is its mass. The composition, classified as X (Hydrogen mass fraction), Y (Helium mass fraction), and Z (Metallicity), is the second most important parameter, although not nearly as relevant as the mass (Hansen et al. 2004). Knowledge of the mass and composition alone can provide astronomers crucial information about the star, such as its total lifetime, which stages of fuel burning occur in the star and where they occur, the core and effective temperatures, among others, as well as how all these values change throughout the star's lifetime.

The Modules for Experiments in Stellar Astrophysics (MESA) (Paxton et al. 2011, 2013, 2015), is a well known stellar model that has been used for the past few years to study stellar interiors of properties by taking user inputs of stellar mass and composition. For this project, the focus is to apply the underlying physics understood to happen in stars to determine the stellar interior properties. These results are then compared with MESA's results. We only analyze stars in the ZAMS (Zero Age Main Sequence) stage of its life in this work. This setting is available in MESA, and it creates a much simpler stellar model compared to more comprehensive ones since the star's evolution is not taken into account.

[1] GitHub Repository available at https://github.com/leonello20/Stellar-Evolution-Project-JHU-2026

2. RELEVANT PHYSICS

2.1. *Stellar Equations*

We start by defining the four governing equations of stellar model. These are differential equations to determine four dependent variables of stellar evolution, which are pressure (P), luminosity (l), radius (r), and temperature (T), as a function of the dependent variable, which is the Lagrangian Coordinate, or the mass (m).

$$\frac{dP}{dm} = -\frac{Gm}{4\pi r^4} \tag{1}$$

$$\frac{dl}{dm} = \epsilon \tag{2}$$

$$\frac{dr}{dm} = \frac{1}{4\pi r^2 \rho} \tag{3}$$

$$\frac{dT}{dm} = -\frac{GmT}{4\pi r^4 P}\nabla \tag{4}$$

Eq. (1) is a representation of hydrostatic equilibrium. That is, this equation is a model for the balance between the gravitational pressure of the star with its radiative pressure. Eq. (2) models the luminosity of the star as a function of its energy generation. We include the proton-proton chain (PP-Chain) reactions as well as the CNO cycle. Eq. (3) is a statement of mass conservation, and Eq. (4) is a model for energy transport. These concepts will be explored in further detail later throughout this section.

In Eq. (4), the ∇ term is given as follows:

$$\nabla = \frac{dlnT}{dlnP} \tag{5}$$

A more useful version of Eq. (5) is the following, as used in this project:

$$\nabla = min\left(\nabla_{rad}, \nabla_{ad}\right) \tag{6}$$

$$\nabla_{rad} = \frac{3}{16\pi ac}\frac{P\kappa}{T^4}\frac{L}{GM} \tag{7}$$

$$\nabla_{ad} = 0.4 \tag{8}$$

The significance of the terms in the equations above will be further explained in Section 2.3.

2.2. *Modeling the energy generation*

In Section 2, we describe the condition that the energy generation mechanism, represented by ϵ was a function of the PP-Chain and the CNO cycle. There are three individual classes of the PP-Chain (PP-I, PP-II, PP-III). The foundational component that makes the PP-Chain possible, however, is the fusion of Hydrogen into Helium. This reaction tends to dominate the energy generation process at a star's lower temperature. Hence, it is dominant in low mass stars. The CNO cycle is characterized as a stellar nucleosynthesis process that uses Carbon, Nitrogen, and Oxygen to fuse four protons into one α particle (^{4}He). Unlike the PP-Chain, this process is responsible for the star's energy generation when at high temperatures. Likewise, the CNO cycle is dominant in massive stars. While not used as part of this project's code, but to show why these statements generally hold, Eq. (9) and Eq. (10) from Hansen et al. (2004) show the approximate relationships for the dependence of the PP-Chain energy generation and the CNO cycle generation, respectively, as a function of density and temperature.

$$\epsilon \approx \epsilon_0 \rho T^4 \tag{9}$$

$$\epsilon \approx \epsilon_0 \rho T^{15} \tag{10}$$

The exponent on the temperature term in for the CNO cycle is often up for debate, and some experts in the field may even use exponents much larger than 15, such as 20, or even up to 40. Furthermore, Kippenhahn et al. (2012) provides equations for how the PP-Chain and CNO cycle as a function of relevant stellar properties, such as temperature and density.

2.3. *Modeling the energy transport*

As introduced in Section 2, we introduced the concept of energy transport. ∇ is a dimensionless temperature gradient. As by Eq. (6), ∇ is always going to be the minimum of the temperature gradients ∇_{rad} and ∇_{ad}. The choice of which of these gradients is used for ∇ highlights which process, radiation or convection, will dominate in transporting energy. This condition determines whether or not the particular star layer is stable to convection, constituting the physics of what is commonly referred to as the Schwarzschild Criterion (Hansen et al. 2004). The criterion is synthesized in Eq. (11) and Eq. (12).

$$\text{If } \nabla_{rad} \leq \nabla_{ad}\text{: transport is via radiation} \tag{11}$$

$$\text{If } \nabla_{rad} > \nabla_{ad}\text{: transport is via convection} \tag{12}$$

We also note that ∇_{rad} depends on κ. This is the Rosseland mean opacity, which depends on the density, temperature, and chemical composition. While Hansen et al. (2004) describes certain approximations for the opacity under certain conditions as a function of density and temperature, we employ pre-generated grids for mean opacity from the Opacity Project (The Opacity Project Team 1995; Seaton 1987). This method ensures that our calculations be consistent with observation. We first determine which grid to use based on the composition, and we then use a Python routine to interpolate between the different density and temperature values that we read from our stellar model.

2.4. *Equation of State*

The Equation of State (EoS) describes how to relate the pressure to the temperature and density. There our two components that determine the total pressure: Ideal Gas Law and Radiation Pressure, as described below:

$$P_{total}(\rho, T) = P_g + P_{rad} \tag{13}$$

$$P_g(\rho, T) = \frac{\rho k T}{\mu m_H} \tag{14}$$

$$P_{rad}(\rho, T) = \frac{1}{3} a T^4 \tag{15}$$

These equations allow us to solve for density as a function of temperature and total pressure:

$$\rho = \frac{\mu m_H}{kT}\left(P_{total} - \frac{1}{3} a T^4\right) \tag{16}$$

From 2.3, we now use the density from Eq. (16) to calculate the opacity from the opacity tables.

3. SOLVING THE STELLAR MODEL

Since we require solving the differential equations described in Section 2.1, we first need to define the boundary conditions. This is similar to any boundary condition problem. The difference now is that we have to consider both the inner boundary conditions as well as the outer boundary conditions of the star. As described later in Section 3.2, we use numerical methods to propagate the boundary conditions to a point inside the star, both from the outside and inside, to find the star's state (pressure, luminosity, total radius, temperature) of the star. This point is chosen such that the stellar mass radially enclosed within this point and the star's center is equal to the star's mass outside this point. If the differences in the state quantities between the inward and outward integrations is small, we have approached a converged solution and the numerical process is complete. However, if the difference is large, we need to correct the boundary conditions. We present algorithms in Section 3.1 that describe an iterative process that propagates the state boundary conditions and modifies them to be used again by the numerical methods until the converged solution is reached. How to determine the validity of the answer is explored in Section 4.

3.1. *Boundary Conditions*

We provide the algorithms introduced above, which allow us to update the state variables whenever we fail to find a converged solution. Our algorithms apply to either the inner or outer boundary conditions. The outward boundary conditions apply as follows to the pressure (Eq. (17)) and temperature (Eq. (18) for the radiative case and Eq. (19) for the convective case):

$$P = P_c - \frac{3G}{8\pi}\left(\frac{4\pi}{3}\rho_c\right)^{4/3} M_r^{2/3} \tag{17}$$

$$T^4 = T_c^4 - \frac{1}{2ac}\left(\frac{3}{4\pi}\right)^{2/3} \kappa \epsilon \rho^{4/3} M_r^{2/3} \tag{18}$$

$$lnT = lnT_c - \left(\frac{\pi}{6}\right)^{1/3} G \frac{\nabla_{ad}\rho_c^{4/3}}{P_c} M_r^{2/3} \tag{19}$$

The inward boundary conditions require conditions at the surface of the star to start with so that we can propagate towards the star's core. We start by writing Eq. (1) in terms of the radial coordinate r by using Eq. (3):

$$\frac{dP}{dr} = -\frac{GM_r}{r^2}\rho = -g\rho \tag{20}$$

We integrate Eq. (20) with respect to r to solve for the pressure at a radius of $r = R_*$:

$$P(r = R_*) = \int_{r=R_*}^{r=\infty} g\rho dr = g_0 \int_{r=R_*}^{r=\infty} \rho dr \tag{21}$$

The term g_0 from Eq. (21) is the gravitational field at the surface of the star, given as follows:

$$g_0 = \frac{GM_*}{R_*^2}\frac{\tau}{\overline{\kappa}} \tag{22}$$

From Eq. (23), τ is the optical depth, given as follows:

$$\tau = \int_{r=R_*}^{r=\infty} \kappa\rho dr = \overline{\kappa}\int_{r=R_*}^{r=\infty} \rho dr = \frac{2}{3} \tag{23}$$

We also write a function to update the luminosity based on the Stefan-Boltzmann Law, which describes how thermal radiation is emitted by an object of a given area at a certain temperature:

$$L = 4\pi R_*^2 \sigma T_{eff}^4 \tag{24}$$

With the boundary conditions established, we explore the numerical solver method used to propagate these quantities through a differential equation solver to obtain a converged solution in Section 3.2.

3.2. *Numerical Methods*

We use the Runge-Kutta method or Order 4 from Python's `scipy.integrate.solve_ivp` to solve the differential equations from 2.1. To determine the number of iterations required to achieve convergence, we compare the results of the outward and inward integrations from the `solve_ivp` and run these against `scipy.optimize.root` algorithm. The `root` function updates the boundary conditions every time the difference between the two solutions is large before running `solve_ivp` again. This completes one iteration. This process repeats until the difference is small enough that convergence is reached.

4. MESA COMPARISON WITH RESULTS

Having completed Section 3.2, we now have all the relevant parameters of the stellar interior and are now able to run these against MESA. We can run this code by substituting in the mass of the star (5 $M_\odot$) and its chemical composition ($X = 0.70$, $Y = 0.28$, $Z = 0.02$) and set a parameter in the code to only consider the ZAMS solution to avoid considering other periods during the star's evolution.

We present in Figure 1 a graphical representation of four of the key stellar quantities analyzed in this model: temperature, pressure, density, and luminosity. We plot these as a function of the Lagrangian mass coordinate M/M_*. For each plot, we include both the results from the model as well as those from MESA. We can see from the graphs that the general trends between the model and MESA plots match fairly well. We also note that the temperature, pressure, and density profiles decrease with the mass coordinate in a very similar fashion to one another. On the other hand, we see that the luminosity increases as the mass coordinate increases.

We also present Table 1, where we compare the luminosity (L), stellar radius (R), effective temperature (T_{eff}), and surface gravity (g) to quantitatively compare the results of these stellar variables between the model and MESA. These variables apply to the surface of the star. We see that all of these fall to within 10% error of MESA's results, which the exception of the luminosity, which falls to within 25% of this ground truth.

5. ENERGY TRANSPORT ANALYSIS

We now turn back to the model results to analyze the energy transport mechanisms occurring within the star. We compare ∇_{ad} and ∇_{rad} to determine which zones are convective and which are radiative. More information on the physics of energy transport is found in Section 2.3. We present a plot in Figure 2 that allows us to compare these two energy mechanisms.

We see that $\nabla_{rad} > \nabla_{ad}$ everywhere from the star's core up until the mass reaches $M/M_* = 0.2312$. This condition flips at this cutoff and stays that way until the surface of the star is reached. From the information presented in Section 2.3, this implies that the transport is convective near the core and radiative towards the envelope. This behavior is expected for a 5 $M_\odot$ star.

6. DISCUSSION

We presented in Section 4 the agreement between the stellar luminosity, radius, effective temperature, and surface gravity. We saw that the radius, effective temperature, and surface gravity were in good agreement with MESA, which may come from a few place. First, we have a 5 $M_\odot$ star, which is massive enough that the effects of electron degeneracy pressure do not significantly impact the Equation of State introduced in Section 2.4. Additionally, since MESA provides hydrostatic models for stars while neglecting rotation and magnetic fields, we suspect that the close agreement between our stellar radius and surface gravity from Section 4 is due to the fact that we ignored those effects as well.

We next look at the luminosity and why its the model's error with respect to MESA's calculation was much greater than that of the other variables. We noted in Section 2.2 that the CNO cycle was predominant in massive stars, such as our own 5 $M_\odot$. From Eq. (10), we see that the energy generation for the CNO cycle is proportional to T^{15} as opposed to an analogous formula for the PP-Chains, which state a proportionality of around T^4. Because we have a massive star, our energy generation is very sensitive to a high change in temperature, which carries a very large exponent. Therefore, even small errors in the temperature may yield large errors in the energy generation. By Eq. (2), we see that the mass derivative of the luminosity directly equals the energy generation. We therefore conclude that the relatively large deviation in luminosity with respect to MESA is highly linked to even fairly small errors in temperature.

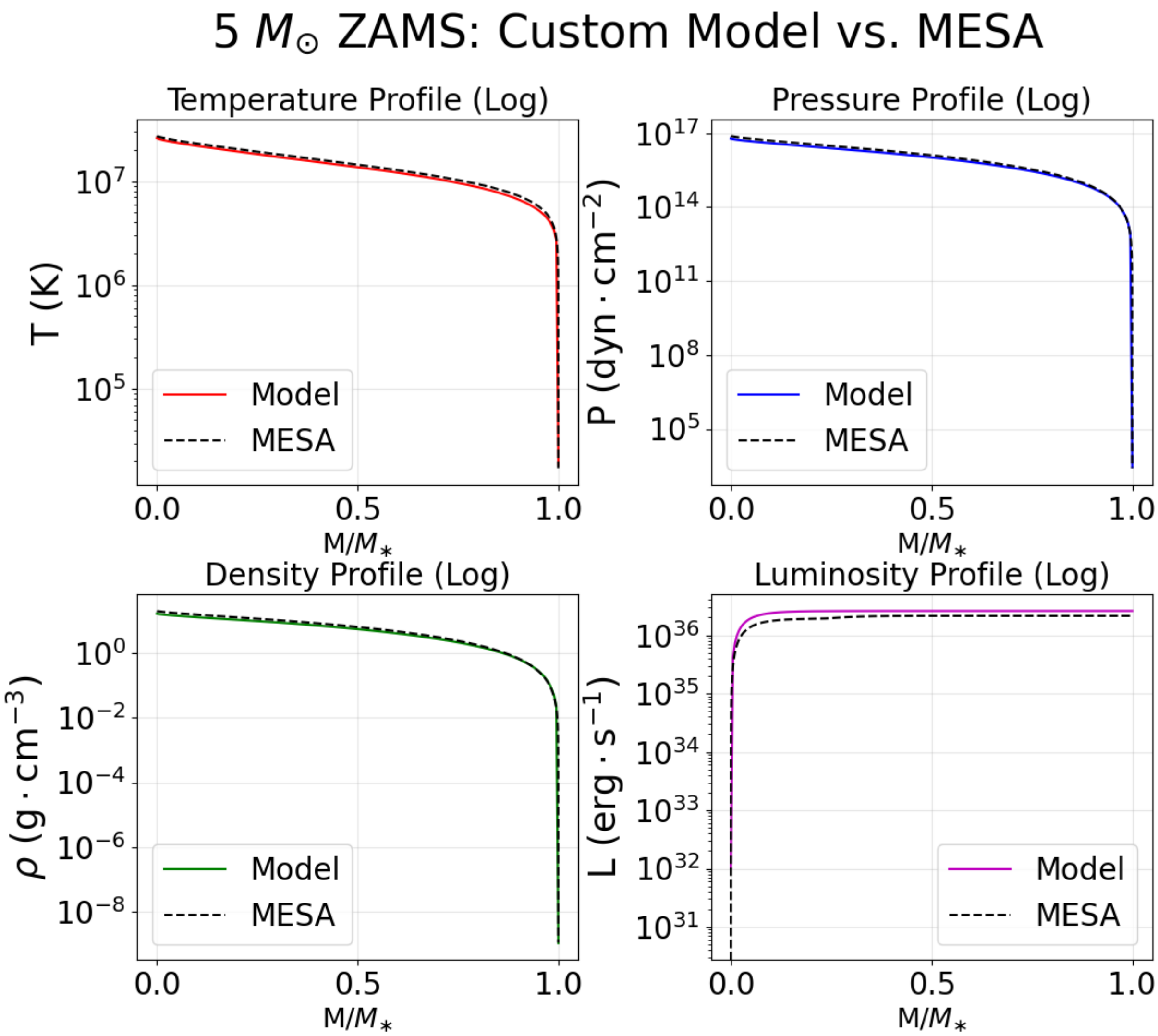

Figure 1. Graph of Model and MESA Results

Table 1. Comparison of Model Results with MESA at Stellar Surface

Parameter	**Model**	**MESA**	**Error (%)**
L ($L_\odot$)	677.72	558.03	21.449
R ($R_\odot$)	2.5530	2.6669	4.2717
$T_{\rm eff}$ (K)	18,432	17,179	7.2948
g ($cm \cdot s^{-2}$)	21,040	19,281	9.1239

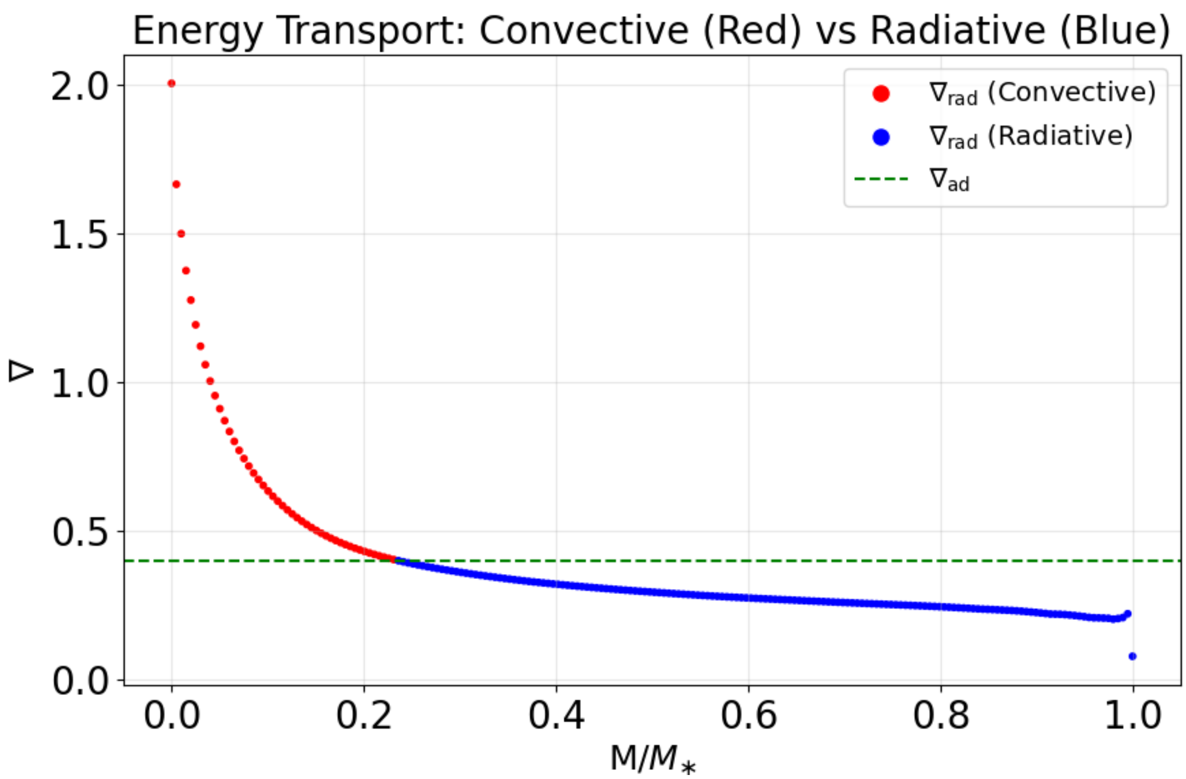

Figure 2. Energy Transport

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